

Energy Efficiency, Cost-Saving Opportunities and Nearly Zero Emissions Analysis in the Residential Sector

Omata David Omakoji^{1,a*}, Opoku Richard^{2,b} and Zinsou Gil-Christ^{3,c}

¹Kwame Nkrumah University of Science and Technology Kumasi, Ghana

^aomatadavid.omakoji@gmail.com*, ^brichard.opoku.richardopoku@gmail.com,

^cygzinsou@gmail.com

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Abstract. This study evaluates the energy efficiency, cost-saving opportunities and nearly zero emissions for the residential sectors of Ghana and Nigeria, focusing on three household appliances: fridges, freezers and bulbs. It analyses the integration of renewable energy systems in residential households, comparing energy-efficient and non-energy-efficient scenarios in both countries. The study uses field data, existing data on RETScreen software and analytical methods to examine the energy consumption patterns of key household appliances and their impact on renewable energy system design, cost, and emissions reduction.

As for the energy consumption patterns, Ghana showed higher daily usage of household appliances, with longer operational hours for fridges and freezers averaging 20.3 hours/day, while Nigeria had lower daily usage, averaging 11.8 hours/day for similar appliances. For Nigeria, the annual electricity cost in the non-energy-efficient scenario is approximately 109.69 USD, while for energy-efficient households, the cost is reduced to 79.31 USD. In Ghana, non-energy-efficient households spend around 379.97 USD annually, compared to 317.55 USD for energy-efficient homes. The results highlight the significant cost-saving opportunities of adopting energy-efficient technologies. This higher energy demand in Ghana and its higher electricity tariffs lead to greater overall consumption and higher costs. Despite similar appliance wattages, Nigeria's lower tariff results in comparatively lower energy expenses.

Using the energy consumption patterns for both countries under the energy-efficient and non-energy-efficient scenarios, system sizing for solar PV and battery storage was conducted to know the economic viability of renewable energy integration through Levelized Cost of Energy (LCOE) and Net Present Value (NPV) assessments. In both scenarios, the payback period for solar PV and battery systems in Nigeria is 22 years, making it not economically viable under current electricity tariffs. In contrast, Ghana shows a payback period of 10.3 years, making solar PV systems financially viable. The RETScreen simulation examined two important scenarios for energy efficiency in Nigeria and Ghana: compact fluorescent lamps (CFLs) as the baseline and LED lighting as a proposed alternative. The results show different GHG (greenhouse gas) reduction equivalences for the number of automobiles that are not driven, the number of individuals who cut their energy use, the number of hectares of forests that absorb GHG, etc.

The study's conclusions highlight the significance of energy efficiency in lowering overall energy usage, tariff rates, and expenses associated with solar and battery systems. The results have also provided more insights for both countries to create more comprehensive policies that will encourage adopting energy-efficient practices and make it cheaper for homes to integrate and use renewable energy systems.

Introduction

The residential sector represents a major share of global energy consumption and carbon emissions. Thus, enhancing energy efficiency and reducing emissions is necessary to check climate change and sustainability (IPCC, 2018). Also, the United Nations Environment Programme (UNEP) states that the climate target of bridging the emissions gap depends on addressing emissions from the residential sector (UNEP, 2017).

Policy frameworks at national and international levels have supported efforts to improve energy efficiency in residential buildings. Boardman (2019) has emphasized energy efficiency policies and their effect on residential buildings and underlined the necessity of devising effective strategies that can achieve the reduction in environmental impacts and energy consumption.

The International Energy Agency (IEA) has also identified potentials related to improvements in energy efficiency regarding residential air-conditioning systems on technology breakthroughs and policy measures (IEA, 2019). In another example, the European Commission describes a scenario in which Nearly Zero-Energy Buildings (nZEBs) are achieved by using existing technologies and processes available in the market (European Commission, 2017).

The lifestyle and habits of the dwellers are equally important to consider regarding the behavioural factors related to the residents' energy consumption patterns. (Shove, 2017).

The United States Department of Energy has been at the forefront of driving energy efficiency initiatives and amplifying activities relating to research and development in residential buildings (US Department of Energy, 2019). This resulted in the best innovative technologies and practices in the field of energy performance enhancement and a decrease in environmental impact.

Literature Review

The residential sectors of Nigeria and Ghana face significant challenges in achieving energy efficiency, realizing cost-saving opportunities, and transitioning toward nearly zero emissions (Ajayi et al., 2018; Adaramola and Paul, 2019). Despite global recognition of the importance of energy efficiency, there is a notable lack of research specifically focused on the unique challenges faced by these two countries. Rapid urbanization, demographic changes, and increasing energy demand stress the urgency of addressing these issues to ensure sustainable development.

Recent studies, such as those by Adamu et al. (2020) and Opoku et al. (2019), primarily examine energy consumption in public and commercial facilities or focus on single appliances like air conditioners. These studies do not comprehensively analyze energy consumption and emissions specific to residential buildings in Nigeria and Ghana. Additionally, existing literature often overlooks the detailed examination of energy consumption patterns at the appliance level within households.

This study addresses the gap by focusing on the residential sector, analyzing the energy-saving and cost-saving potential of adopting energy-efficient appliances, and assessing the potential for nearly zero emissions with renewable energy integration. The research is critical for informing policymakers, urban planners, and stakeholders in both countries, helping them develop tailored interventions that align with local realities and contribute to global efforts to mitigate climate change.

Methods

The research design for this study adopts a mixed-method approach, integrating both qualitative and quantitative methods to thoroughly address the research objectives. The study begins with a comprehensive survey designed to collect primary data on household energy consumption, appliance usage patterns, and emissions from a stratified random sample of households in various regions of Ghana and Nigeria. The survey gathers detailed information on the types and number of appliances, their usage duration and frequency, monthly energy bills, and energy sources, alongside demographic data. The primary data is complemented by secondary data sourced from national energy statistics, utility companies, and relevant governmental and non-governmental reports. Descriptive statistics and comparative analysis were used to analyze the collected data, identifying key trends and patterns in energy consumption, tariffs and emissions.

The analysis involved a cost-benefit analysis to evaluate energy savings from adopting energy-efficient appliances. Key metrics like initial costs, energy consumption, and life cycle costs were analyzed for both conventional and energy-efficient appliances. The differences in energy consumption between the traditional and energy-efficient devices were used to calculate potential savings in electricity costs. Regression analysis was applied to assess the relationship between

appliance efficiency and overall household energy consumption. Additionally, correlation analysis was conducted using Python to evaluate relationships between household types and energy-consuming appliances like bulbs, refrigerators, and freezers.

Data Collection

Data collection methods were implemented according to the research design. Surveys and questionnaires were administered to gather quantitative data on energy consumption, appliance usage, and consumer preferences.

Quantitative Analysis

Quantitative data collected from surveys were analysed using statistical techniques including Microsoft excel, power BI and python. Descriptive statistics, regression and correlation analysis, and cost-benefit analysis were used to quantify energy savings, cost reductions, and emissions reductions associated with energy-efficient appliances.

Objective 1: Assess the current state of energy consumption, appliance usage pattern, and emissions in the residential sectors of Ghana and Nigeria.

To achieve this objective, researchers employ mathematical formulas to quantify energy consumption, appliance usage patterns, and emissions, as shown below:

$$(1). \text{ Average Daily Energy Consumption} = \frac{\sum_{i=1}^n E_i}{n} \dots\dots\dots \text{Eq. 1}$$

Where:

- E_i is the daily energy consumption of the i -th household for the specific type of bulb (measured in watt-hours per day).
- n is the total number of households using that type of bulb.

In this context, E_i can be calculated for each household as:

$$(2). E_i = \sum_{j=1}^m (W_j \times H_j) \dots\dots\dots \text{Eq. 2}$$

Where:

- W_j is the wattage of the j -th bulb
- H_j is the number of hours the j -th bulb is used per day.
- m is the total number of bulbs of that type in the i -th household.

Combining these formulas, the overall average daily energy consumption for each type of bulb is:

$$(3). \text{ Average Daily Energy Consumption} = \frac{\sum_{i=1}^n \sum_{j=1}^m (W_j \times H_j)}{n} \dots\dots\dots \text{Eq. 3}$$

Objective 2: Evaluate the potential benefits of implementing higher energy-efficient appliances for energy savings, cost reduction, and emissions reduction.

To analyze the cost savings from using energy-efficient refrigerators, the researcher utilized the average electricity consumption data for both energy-efficient and non-energy-efficient refrigerators.

- C_{ne} = Cost of electricity for non-energy-efficient refrigerator
- C_e = Cost of electricity for an energy-efficient refrigerator
- E_{ne} = Daily energy consumption of non-energy-efficient refrigerator (in watt-hours)
- E_e = Daily energy consumption of energy-efficient refrigerator (in watt-hours)
- T = Electricity tariff (cost per kWh)
- M = Number of days in a month (typically 30)
- Y = Number of months in a year (12)

Daily Consumption Calculation:

$$(4). \text{ Daily Consumption (kWh)} = \frac{E}{1000} \dots\dots \text{Eq. 4}$$

(I.e. the energy consumption Wh divided by 1000 to convert to kWh)

Monthly Consumption:

$$(5). \text{ Monthly Consumption (kWh)} = \text{Daily Consumption (kWh)} \times M \dots\dots\dots \text{Eq. 5}$$

Annual Consumption:

$$(6). \text{ Annual Consumption (kWh)} = \text{Monthly Consumption (kWh)} \times Y. \dots\dots\dots \text{Eq. 6}$$

Cost Calculation:

$$(7). C_{\text{monthly}} = \text{Monthly Consumption (kWh)} \times T \dots\dots\dots \text{Eq. 7}$$

$$(8). C_{\text{Annual}} = \text{Annual Consumption (kWh)} \times T \dots\dots\dots \text{Eq. 8}$$

Cost Savings:

$$(9). \text{ Monthly Cost Saving} = C_{\text{ne,monthly}} - C_{\text{e,monthly}}. \dots\dots\dots \text{Eq. 9}$$

$$(10). \text{ Annual Cost Saving} = C_{\text{ne,annual}} - C_{\text{e,annual}} \dots\dots\dots \text{Eq. 10}$$

Techno-Economic Analysis of the Renewable Energy Integration

The daily energy consumption (in kWh/day) for Nigeria and Ghana across two scenarios: non-energy-efficient (using a non-energy-efficient fridge, freezer, and CFL bulbs) and energy-efficient (using an energy-efficient fridge, freezer, and LED bulbs) were calculated from which the daily electricity cost for the appliances was calculated using the countries different tariff systems.

Using the daily energy consumption for both energy-efficient and non-energy-efficient scenarios in both countries, the researcher calculated the annual energy consumption and cost based on the respective electricity tariffs.

$$(11). \text{ Annual Cost (USD)} = \text{Daily Energy Consumption (kWh/day)} \times 365 \times \text{Tariff (USD/kWh)} \dots\dots \text{Eq. 11}$$

Based on the daily energy consumption for each scenario, the researcher designed solar PV systems capable of providing that amount of energy, including battery storage for backup during nighttime or low sunlight hours.

(12). Solar PV Sizing Formula:

$$\text{Solar PV Size (kW)} = \frac{\text{Daily Energy Consumption (kWh/day)}}{\text{Peak Sunlight Hours } \left(\frac{\text{hours}}{\text{day}}\right)} \dots\dots\dots \text{Eq. 12}$$

(13). Battery Sizing Formula:

$$\text{Battery Size (kWh)} = \frac{\text{Daily Energy Consumption (kWh/day)}}{2} \dots\dots\dots \text{Eq. 13}$$

The LCOE for both scenarios in each country was calculated and compared, showing the long-term cost-effectiveness of energy-efficient technology when paired with renewable energy systems.

$$(14). \text{ LCOE} = \frac{\text{Total System Cost (USD)}}{\text{Total Energy Generated Over Lifetime (kWh)}} \dots\dots\dots \text{Eq. 14}$$

The net present value for both scenarios in each country was calculated and compared to determine the economic viability of household renewable adoption.

$$(15). \text{ NPV} = \sum \left(\frac{\text{Annual Savings}}{(1+r)^t} \right) - \text{Total System Cost} \dots\dots\dots \text{Eq. 15}$$

Where $r = 0.05$ (5% discount rate), $t = \text{year}$

Lifespan: 20 years

Objective 3: Analyse the energy management approach to realize nearly zero emissions in the residential sector.

Emission analysis considered two scenarios: one for a 200m² building that uses compact fluorescent lights, which were used as the base case, and another for the same building type that uses LED lights. The analysis was conducted using the RETScreen application using the scenario assumption, as shown in table 1. To successfully run this scenario, the climate data for each location were calculated using equations 16 and 17 below.

Table 1. General assumptions for the scenario analysis

General assumptions	
Square meter of building used	200m ²
Operating hours for the bulbs	10hours/day
TandD losses for Nigeria	7.5%
TandD losses for Ghana	3.7%
The Base case for the bulb	
Type of bulb	Compact fluorescent light
Manufacturer	GE
Model	GE A19 – 2700k,15w
Number of lamps per fixture	1
Electricity load per lamp	15w
Electricity load per fixture	15w
Efficiency	44lm/w
Color temperature	2700k
Luminous	660lm
The Proposed case for the bulb	
Type of bulb	Light Emitting Diode (LED)
Manufacturer	GE
Model	GE energy smart A19 – 3000k, 9w
Number of lamps per fixture	1
Electricity load per lamp	9w
Electricity load per fixture	9w
Efficiency	50lm/w
Color temperature	3000k
Luminous	450lm

The climate Data for Ghana and Nigeria was calculated using the formula:

$$(16). I_o = \frac{12.3600}{\pi} \cdot G_{sc} \cdot [1 + 0.033 \cdot \cos \frac{360 \cdot n}{365}] \cdot [\cos \phi \cdot \cos \delta \cdot (\sin \omega_2 - \sin \omega_1) + \frac{\pi(\omega_2 - \omega_1)}{180} \sin \phi \cdot \sin \delta] \quad \text{..Eq. 16}$$

$$(17). H_o = \frac{24.3600}{\pi} \cdot G_{sc} \cdot [1 + 0.033 \cdot \cos \frac{360 \cdot n}{365}] \cdot [\cos \phi \cdot \cos \delta \cdot \sin \omega_s + \frac{\pi \cdot \omega_s}{180} \sin \phi \cdot \sin \delta] \quad \text{.. Eq.17}$$

- Solar constant: G_{sc} [W/m²].
- Hourly radiation: I_o [J/m²]
- Daily radiation: H_o [J/day. M²]

The calculation of the air temperature using solar radiation:

A simple empirical formula to estimate temperature based on solar:

$$(18). \quad T = T_0 + k \times \text{Solar Radiation.} \quad \text{.... Eq. 18}$$

Where:

T = estimated temperature

T₀ = base temperature (intercept term, often a mean temperature)

K = coefficient that relates solar radiation to temperature (derived from historical data)

Solar Radiation = amount of solar energy received per unit area.

Data Analysis Procedures

Descriptive Statistics

Descriptive statistics were employed to summarize and describe the key characteristics of the dataset. Measures such as mean, median, standard deviation, and range were calculated for variables such as energy consumption, appliance usage, and cost of energy/usage. These measures provided an overview of the data's central tendency, dispersion, and distribution.

$$(19). \quad \dots \text{Mean } (\mu): \frac{\sum x_i}{n} \dots \text{Eq. 19}$$

$$(20). \quad \dots \text{Standard Deviation } (\sigma): \dots \text{Eq. 20}$$

$$\sigma = \sqrt{\frac{\sum (x_i - \mu)^2}{N}}$$

σ = population standard deviation

N = the size of the population

x_i = each value from the population

μ = the population mean

Regression Analysis

Regression analysis explored the relationships between variables and identified factors influencing energy consumption patterns and cost implications. Simple linear regression analysis examined the relationship between independent variables and energy-related outcomes.

Simple Linear Regression:

$$(21). \quad Y = \beta_0 + \beta_1 X + \epsilon \dots \text{Eq. 21}$$

Where:

- Y is the dependent variable.
- X is the independent variable.
- β_0 is the intercept (the value of Y when $X=0$).
- β_1 is the slope (the change in Y for a unit change in X).
- ϵ is the error term.

Correlation Analysis

A correlation analysis was run to understand the relationship between the House types, bulbs, refrigerator and Chest freezer. This was further visualized into the correlation heatmap using python.

$$(22). \quad \frac{n\sum XY - \sum X \sum Y}{\sqrt{(n\sum X^2 - (\sum X)^2)(n\sum Y^2 - (\sum Y)^2)}} \dots \text{Eq. 22}$$

The terms in that formula are:

n = the number of data points, i.e (x, y) pairs, in the data set.

$\sum XY$ = the sum of the product of the x-value and y-value for each point in the data set.

$\sum X$ = the sum of the x-values in the data set.

$\sum Y$ = the sum of the y-values in the data set.

$\sum x^2$ = the sum of the squares of the x-values in the data set.

$\sum Y^2$ = the sum of the squares of the y-values in the data set

Results and Discussion

Average Bulb Usage and Number of Bulbs in Each Household Type

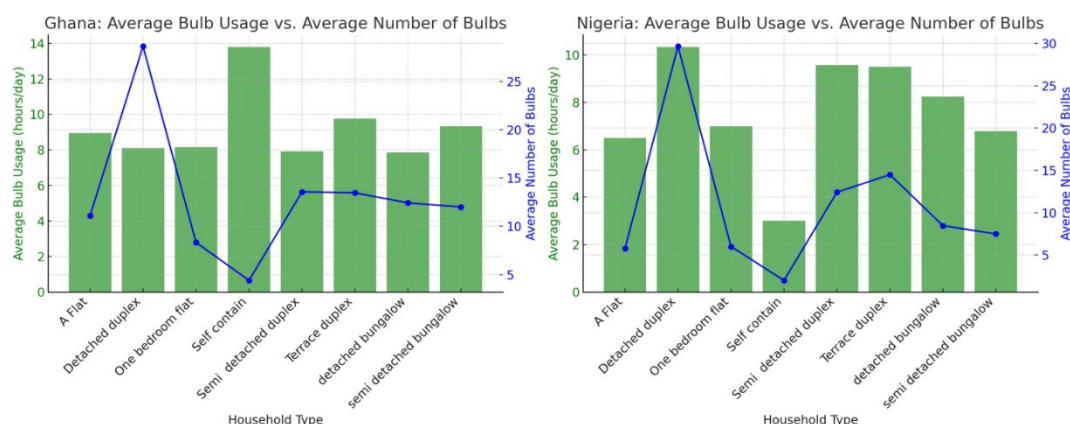


Fig. 1. Average bulb Usage and Number of Bulbs in Each Household Types

The graphs above compare the **average bulb usage**, **average number of bulbs**, and **daily energy consumption of bulbs** for different household types in **Ghana** and **Nigeria**. The green bars represent the average hours per day that bulbs are turned on, while the blue line with markers shows the average number of bulbs in each household.

Average Daily Energy Consumption of the Bulbs for each house

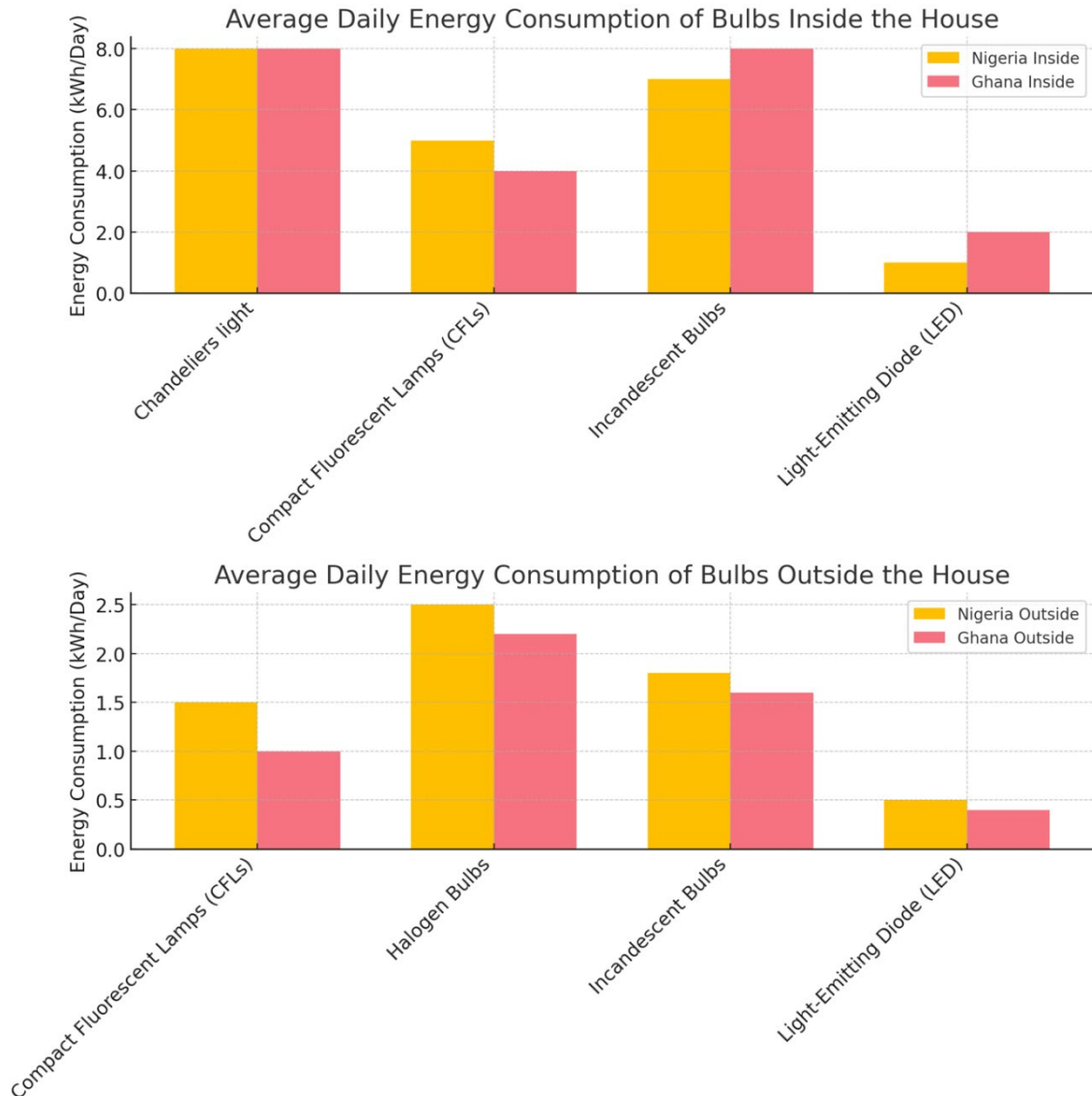
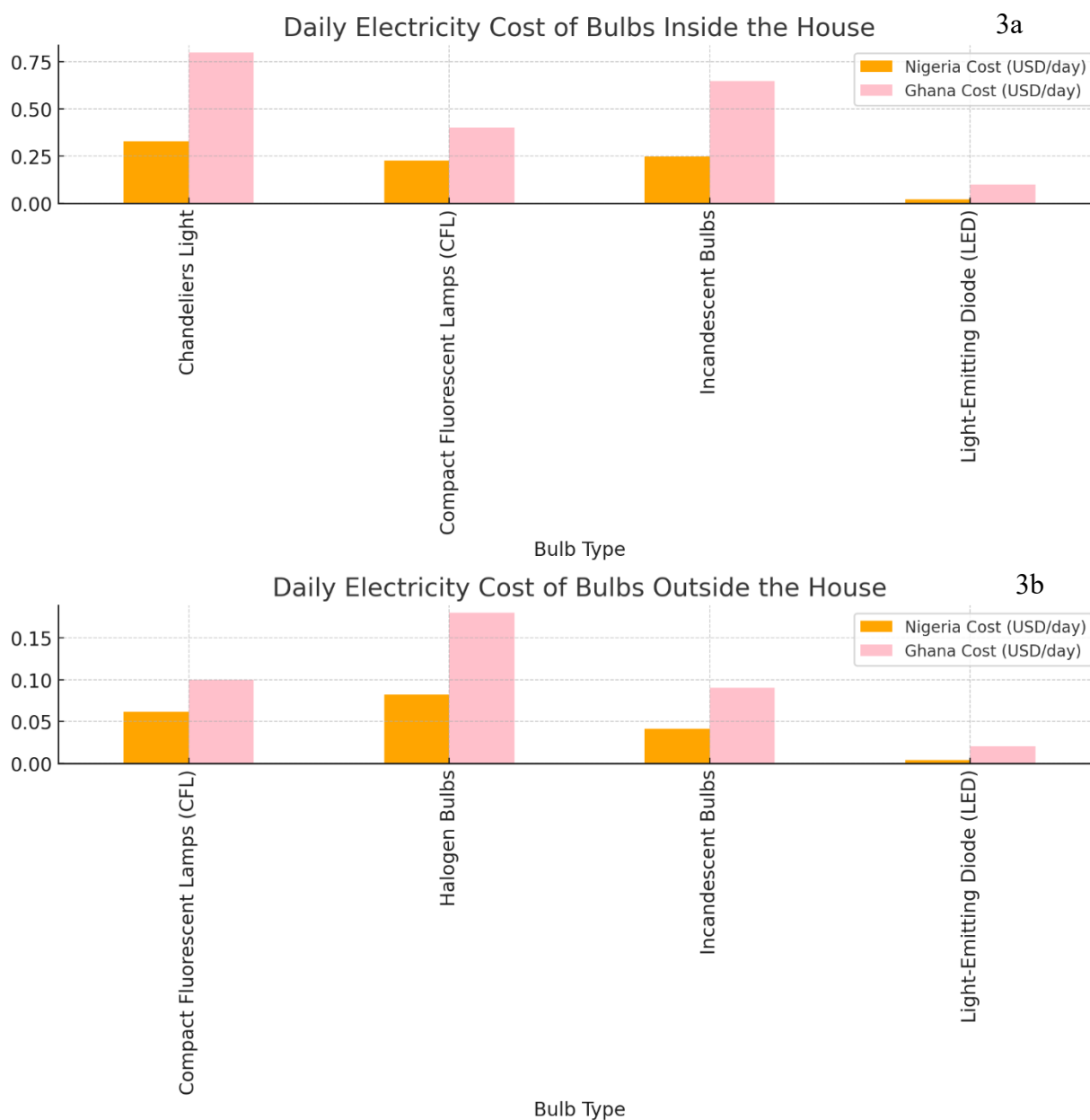


Fig.2 Energy consumption of bulbs within and outside the house

The analysis of the average daily energy consumption for different types of bulbs inside and outside the house reveals significant differences between Nigeria and Ghana. These differences can provide insights into both countries' energy efficiency practices and preferences.

Average Daily Electricity Cost for Each Bulb Type

**Fig. 3a.** Average Daily Electricity Cost for Bulb Inside the House**Fig. 3b.** Average Daily Electricity Cost for Bulb Outside the House

The graphs above depict the daily electricity cost for various types of bulbs used inside and outside the house in Nigeria and Ghana, based on the average respective electricity tariffs at the time the study was conducted (Nigeria: 0.041 USD/kWh and Ghana: 0.1 USD/kWh).

Average Hourly Usage of the Fridge and Freezer

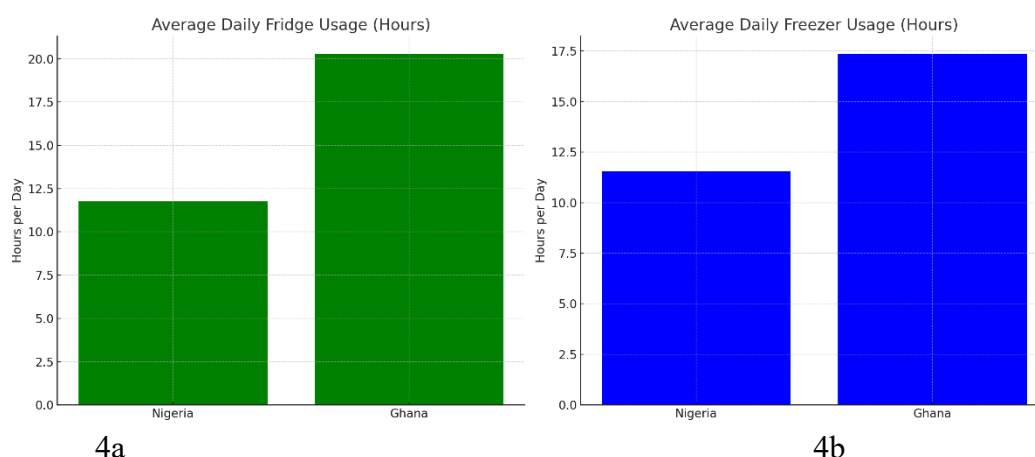


Fig. 4a Average Hourly Usage of the Fridge and Freezer

Fig. 4b Average Hourly Usage of the Fridge and Freezer

In Ghana, refrigerators are turned on for an average of 20.3 hours per day, while in Nigeria, they are on for about 11.8 hours per day. The higher usage in Ghana could be due to factors such as more reliable electricity supply, lifestyle differences, or higher energy efficiency in appliances that allow for prolonged operation. In contrast, Nigerian households may experience frequent power outages, leading to shorter operational hours for refrigerators.

Similarly, freezers in Ghana operate for an average of 17.4 hours per day, compared to 11.6 hours per day in Nigeria. The extended usage of freezers in Ghana indicates a similar trend to refrigerator usage, suggesting that electricity supply reliability allows for longer usage. In Nigeria, the relatively lower freezer usage could be attributed to intermittent electricity access or the prioritization of energy consumption for other essential needs.

Analysis of Average Wattage of the Fridge and Freezers

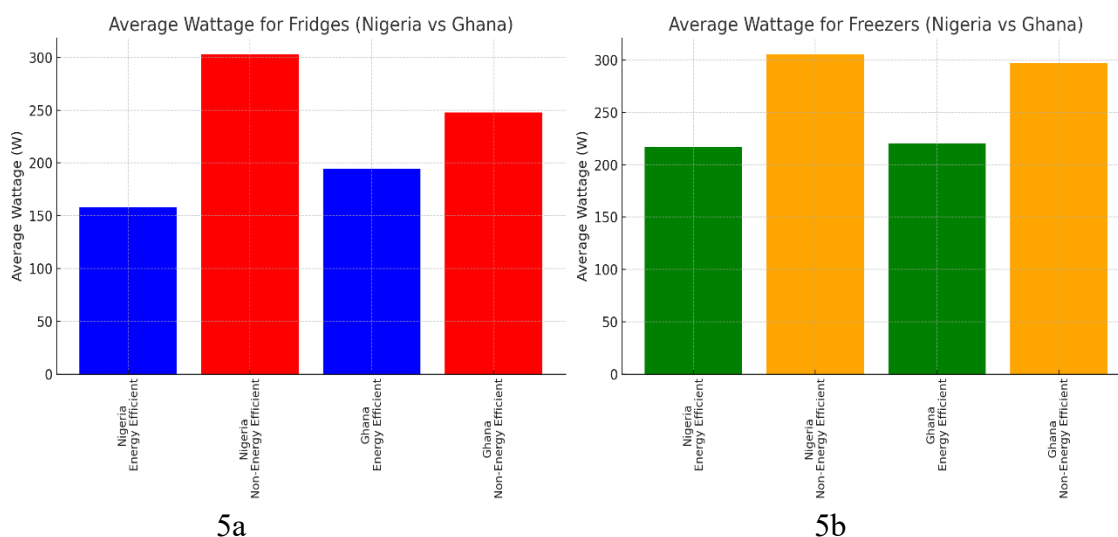


Fig. 5a. Analysis of the average Wattage of Fridges in Nigeria and Ghana

Fig. 5b. Analysis of the average Wattage of Freezers in Nigeria and Ghana

For energy-efficient fridges, the average wattage in Nigeria is 158 watts, while in Ghana it is slightly higher at 194 watts. This indicates that even energy-efficient fridges in Ghana tend to consume more power than in Nigeria, which could be due to differences in appliance models or energy efficiency standards. In the case of non-energy-efficient fridges, the difference is more pronounced. In Nigeria, non-energy-efficient fridges consume 303 watts on average, while the figure is 248 watts in Ghana. This suggests that non-energy-efficient fridges in Nigeria tend to consume significantly more power than those in Ghana.

There is little difference between the two countries for energy-efficient freezers. In Nigeria, energy-efficient freezers consume 216 watts; in Ghana, the figure is 220 watts, indicating similar levels of energy efficiency in freezer models between the two countries. For non-energy-efficient freezers, both countries see a significant increase in power consumption. In Nigeria, these freezers consume 305 watts, while in Ghana, they consume 297 watts. This highlights the energy-saving potential of opting for energy-efficient models.

Average Daily Energy Consumption (KWh) for the Fridge and Freezer

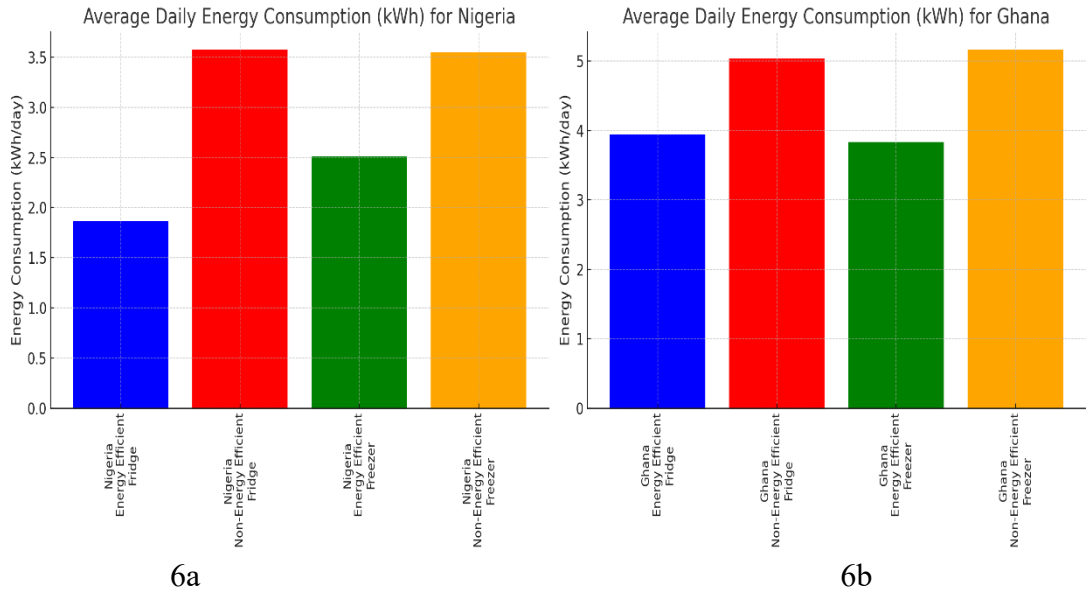


Fig. 6a Average Daily Energy Consumption (KWh) for the Fridge and Freezer in Nigeria
Fig. 6b Average Daily Energy Consumption (KWh) for the Fridge and Freezer in Ghana

In both countries, adopting energy-efficient appliances would help reduce energy consumption, lower electricity bills, and contribute to environmental sustainability. This demonstrates the importance of choosing energy-efficient models for household appliances.

Daily Energy Consumption for Different Scenarios in Nigeria and Ghana

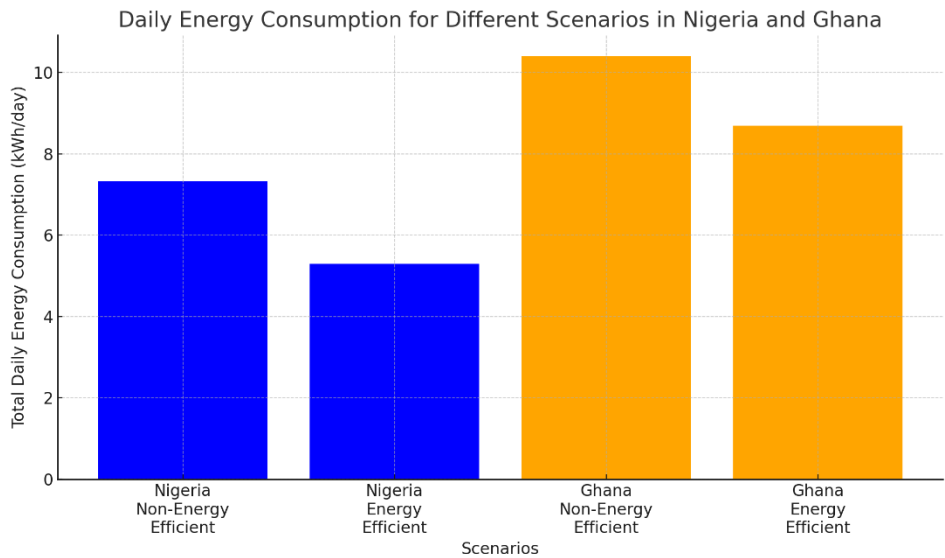


Fig. 7. Daily Energy Consumption for Different Scenarios in Nigeria and Ghana

This graph compares daily average energy consumption (in kWh/day) for Nigeria and Ghana across two scenarios: non-energy-efficient (using a non-energy-efficient fridge, freezer, and CFL bulbs) and energy-efficient (using an energy-efficient fridge, freezer, and LED bulbs).

Regression Analysis for the Number of Households and Unit of Electricity (kwh/month)

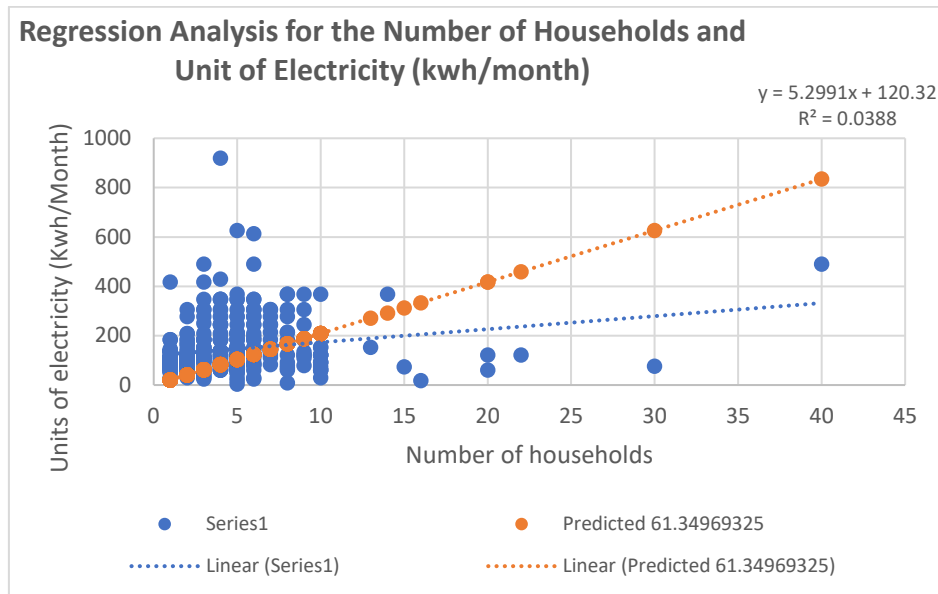


Fig. 8. Regression analysis of the number of households and units of electricity

Table 2. Regression analysis

Multiple R	R Square	Adjusted R Square	Standard Error	Observations	Intercept
0.19805065	0.03922406	0.03664825	100.778805	375	119.995666

The weak correlation and low R Square value suggest that the number of households alone is not a strong predictor of energy consumption. This aligns with findings from other studies, which indicate that while household size does impact energy use, other factors such as the type and efficiency of appliances, income levels, and behavioural patterns play a significant role (O'Neill and Chen, 2002; Jones, Lomas, and Skea, 2010).

Cost Saving Analysis

Daily Electricity Cost for Fridges and Freezers

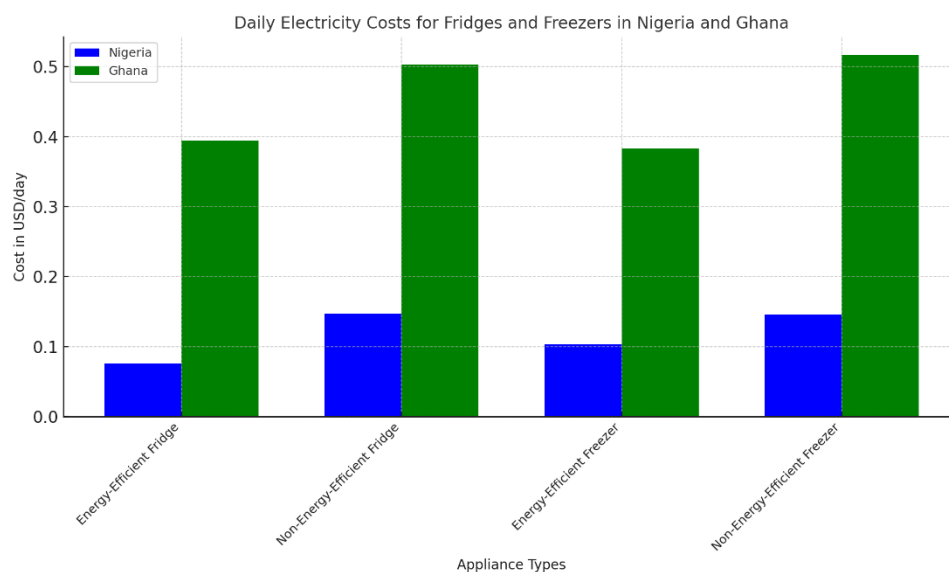


Fig. 9. Daily Electricity Cost for Fridges and Freezers

The graph emphasizes the financial benefits of energy-efficient appliances, with more dramatic savings in countries like Ghana, where electricity tariffs are higher. This makes the case for prioritizing energy efficiency to reduce daily energy costs, especially in regions with elevated electricity prices.

Correlation Analysis for the House types, bulbs, refrigerator and Chest freezer

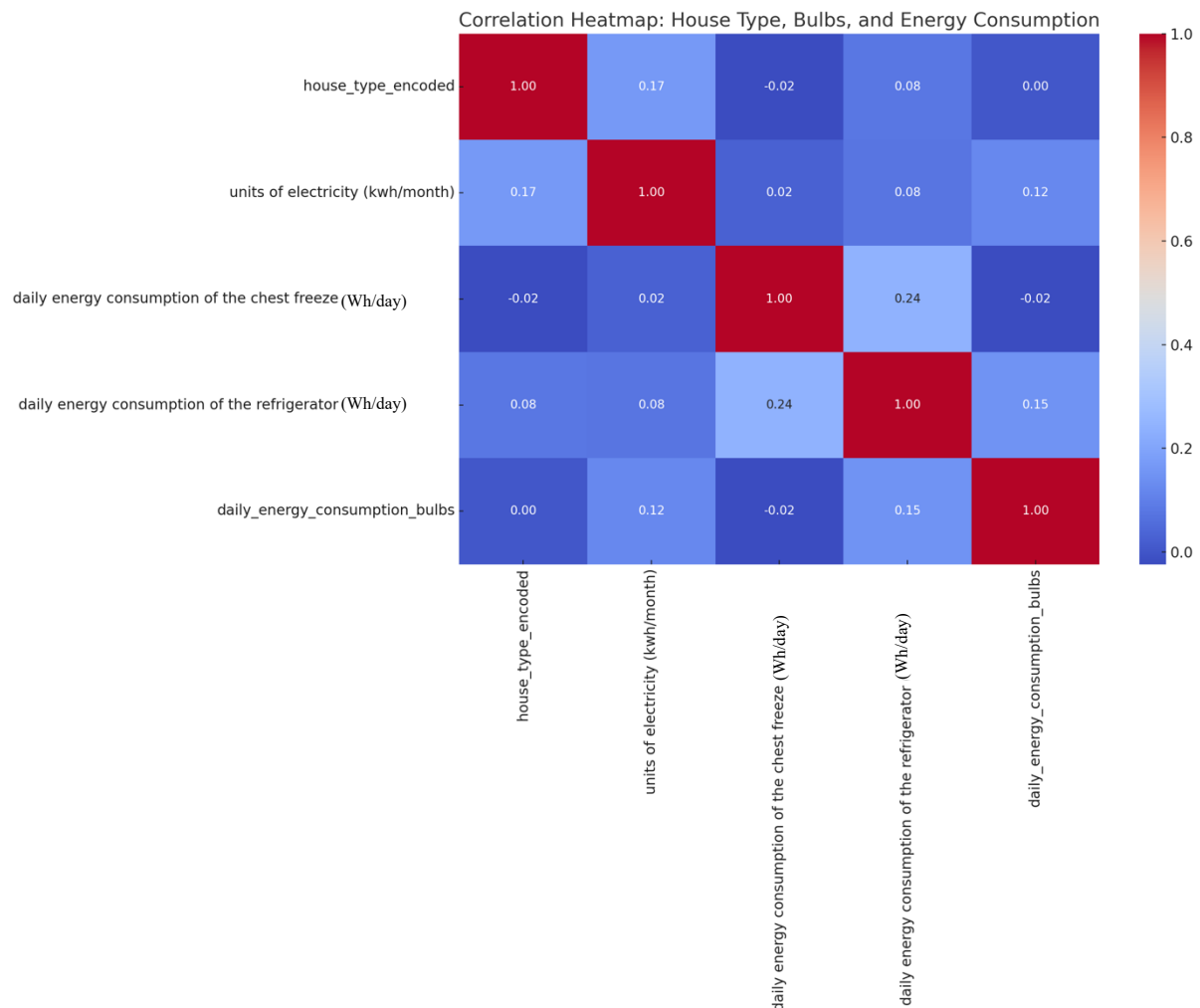


Fig. 10. Heatmap correlation analysis of house types, bulbs, refrigerator and Chest freezer

In the heatmap, the values range from -1 to 1, representing the correlation coefficients. A correlation coefficient of +1 indicates a perfect positive correlation, where an increase in one variable corresponds to an increase in the other. Conversely, a correlation coefficient of -1 indicates a perfect negative correlation, where an increase in one variable corresponds to a decrease in the other. A correlation coefficient of 0 signifies no linear relationship between the variables.

Emission Reduction Analysis Use RETScreen

The emission analysis considered two scenarios: one for a detached single-unit building using compact fluorescent lights, which was used as the base case, and another for a single-unit building with LED lights. The analysis was conducted using the RETScreen application.

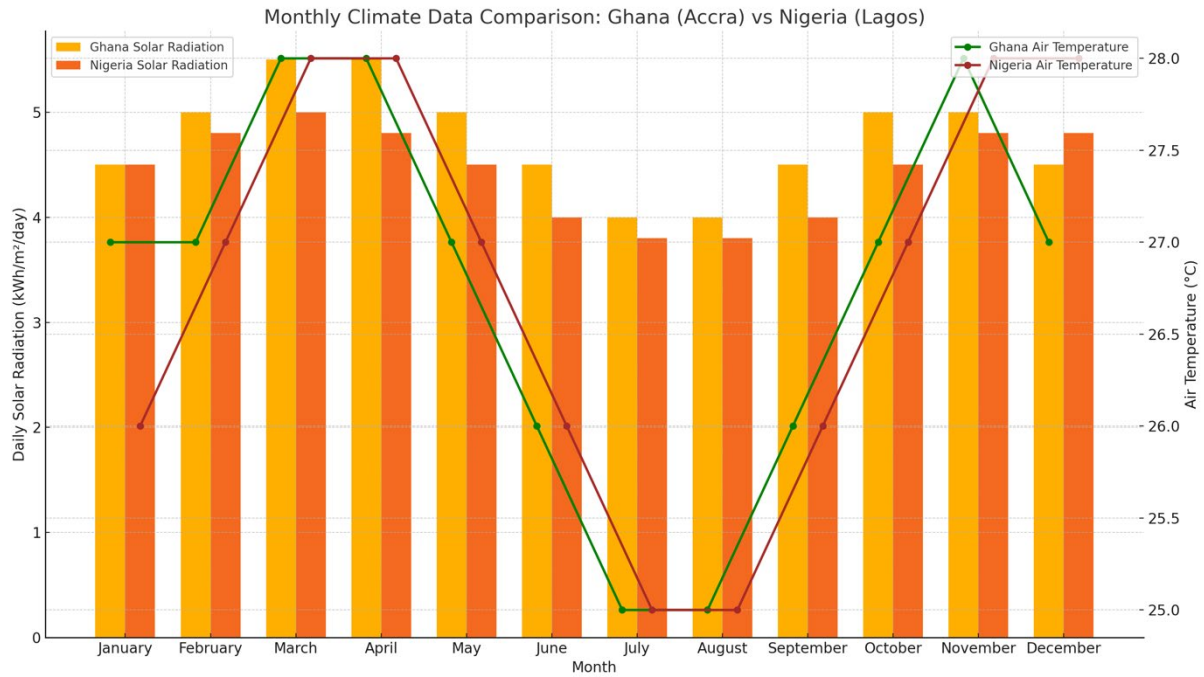


Fig. 11. Monthly climate data comparison

Energy Saving and Fuel Summary

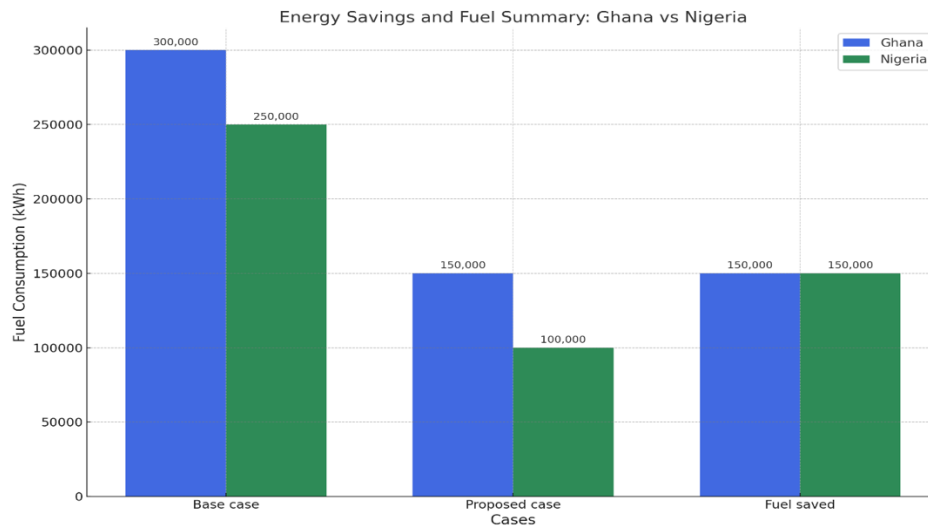


Fig. 12. Energy savings and fuel summary

The graph effectively showcases the potential energy savings that can be achieved by transitioning from CFL to LED lighting. The significant reduction in fuel consumption in Ghana and Nigeria proves the effectiveness of LED technology in reducing energy demand. These findings are consistent with existing literature on the benefits of energy-efficient lighting technologies.

House Gas (GHG) Emission Comparison

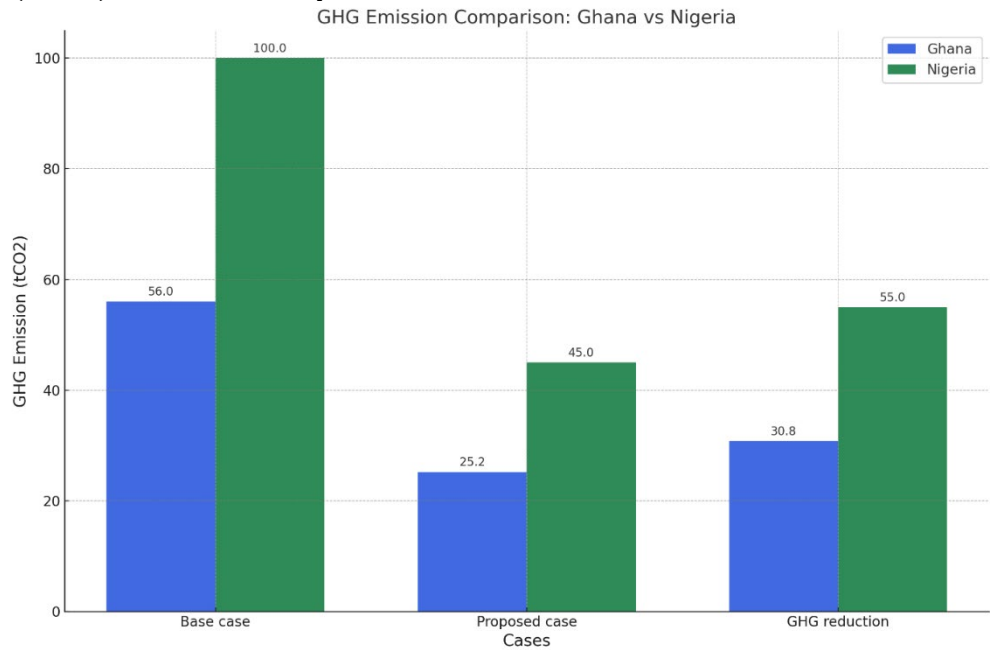


Fig. 13. Greenhouse gas emissions

Comparison of GHG Emission Reduction Equivalence (Liters of Gasoline Not Consumed)

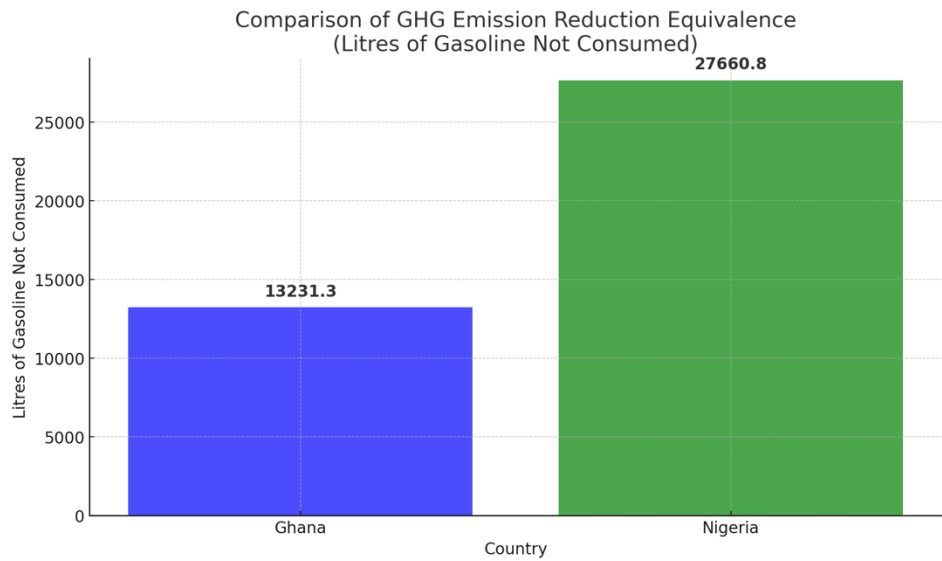


Fig. 14. Greenhouse gas emission reduction equivalences

Comparison of GHG Emission Reduction (Cars and Light Trucks Not Used)

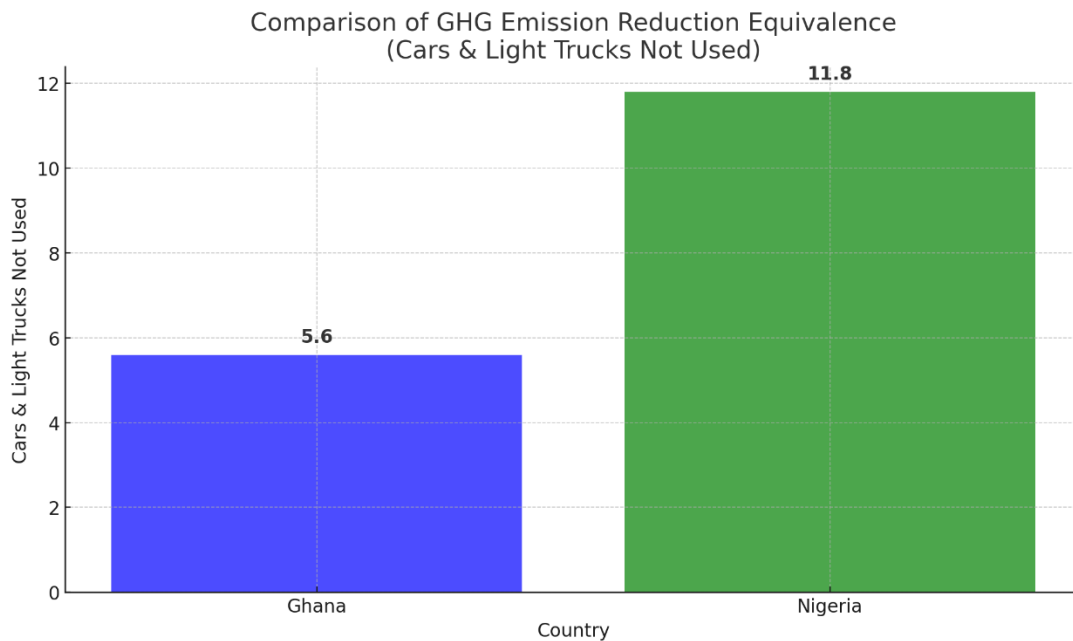


Fig. 15. Comparison of GHG Emission Reduction (Cars and Light Trucks Not Used)

According to the graph, Ghana's reduction in GHG emissions is equivalent to taking approximately 5.6 cars and light trucks off the road. In comparison, Nigeria's reduction is equivalent to about 11.8 cars and light trucks. This comparison highlights the significant environmental impact of adopting energy-efficient technologies, demonstrating how switching to efficient lighting can reduce emissions, similar to removing several vehicles from operation. The data aligns with findings from the U.S. Environmental Protection Agency (EPA), which states that replacing less efficient lighting with LEDs can considerably lower GHG emissions, contributing to climate change mitigation efforts (EPA, 2020).

Comparison of GHG Emission Reduction (Barrels of Crude Oil Not Consumed)

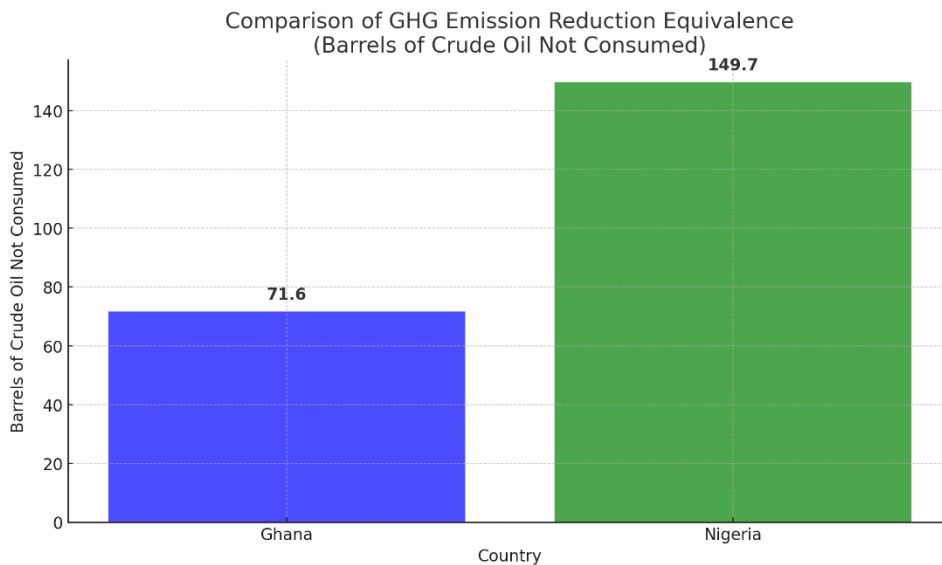


Fig. 16. Comparison of GHG Emission Reduction (Barrels of Crude Oil Not Consumed)

The graph illustrates the environmental benefits of reduced greenhouse gas (GHG) emissions by comparing the equivalent amount of crude oil not consumed in Ghana and Nigeria. The data aligns with the International Energy Agency (IEA) findings, which indicate that energy efficiency

improvements can lead to substantial reductions in oil consumption, thereby contributing to global efforts to mitigate climate change and promote sustainable energy use (IEA, 2019).

Comparison of GHG Emission Reduction Equivalence (Hectares of Forest Absorbing Carbon)

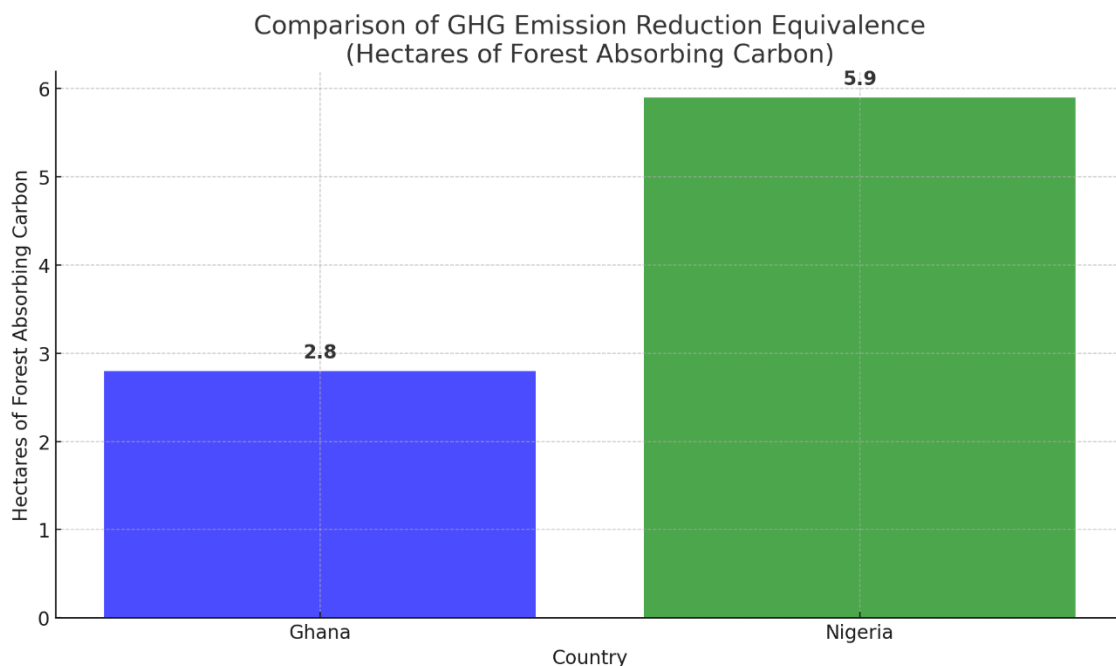


Fig. 17. Comparison of GHG Emission Reduction (Hectares of Forest Absorbing Carbon)

The graph above visualizes the greenhouse gas (GHG) emission reductions from energy-efficient lighting technologies regarding the equivalent amount of forest area required to absorb the same amount of carbon dioxide. According to the Intergovernmental Panel on Climate Change (IPCC), improvements in energy efficiency are essential for reducing emissions and achieving global climate goals. This graph demonstrates the practical implications of such improvements (IPCC, 2014).

Comparison of GHG Emission Reduction Equivalence (People Reducing Energy Use by 20%)

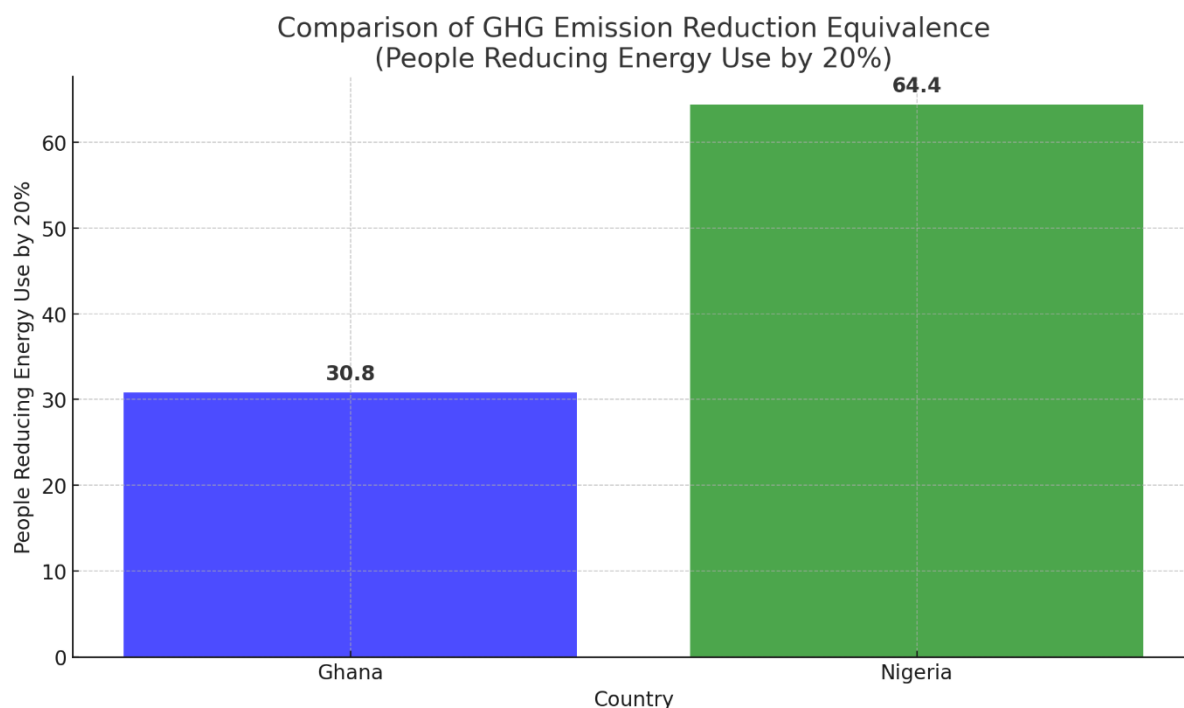


Fig. 18. Comparison of GHG Emission Reduction (People Reducing Energy Use by 20%)

The graph compares the impact of GHG emission reductions achieved through energy-efficient lighting technologies in Ghana and Nigeria to the equivalent number of people reducing their energy use by 20%. Studies, such as those by the International Energy Agency (IEA), support that energy efficiency is one of the most effective strategies for reducing emissions and promoting sustainability (IEA, 2019).

Scenario Analysis, Renewable Energy Integration and Economic Analysis for Ghana and Nigeria.

The annual electricity cost for both energy-efficient and non-energy-efficient scenarios in Nigeria and Ghana was calculated using daily energy consumption data and respective electricity tariffs. The **daily energy consumption** (in kWh/day) for **Nigeria** and **Ghana** across two scenarios: **non-energy-efficient** (using a non-energy-efficient fridge, freezer, and CFL bulbs) and **energy-efficient** (using an energy-efficient fridge, freezer, and LED bulbs) were used to design solar PV systems and battery backup systems to meet daily energy demands, considering backup storage for nighttime. The costs of these systems were calculated using general market prices for solar PV panels and batteries in each country.

The researcher compared the energy-efficient and non-energy-efficient scenarios, showing how adopting energy-efficient appliances reduces the cost of installing solar PV systems. Finally, a comparison of the levelized cost of energy (LCOE) is made to demonstrate the long-term cost-effectiveness of pairing energy-efficient technology with renewable energy solutions, highlighting potential savings and financial benefits over time.

Daily Energy Consumption of Different Scenarios in Nigeria and Ghana

Table 3. Daily Energy Consumption of Different Scenarios in Nigeria and Ghana

Scenario	Total Daily Energy Consumption (kWh/day)
Non-Energy Efficient (Fridge + Freezer + CFL) - Nigeria	7.330223895
Energy Efficient (Fridge + Freezer + LED) - Nigeria	5.295419401
Non-Energy Efficient (Fridge + Freezer + CFL) - Ghana	10.40678389
Energy Efficient (Fridge + Freezer + LED) - Ghana	8.696839401

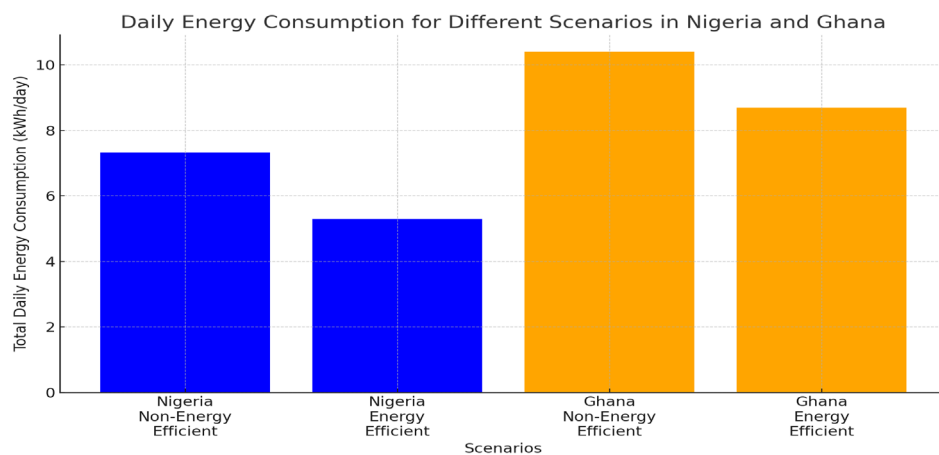


Fig. 19. Daily Energy Consumption of Different Scenarios in Nigeria and Ghana

The graph above shows the daily energy consumption for both energy-efficient and non-energy-efficient scenarios in Nigeria and Ghana, with Ghana exhibiting higher consumption in both cases. A significant factor driving this disparity is the difference in electricity tariffs. Ghana's tariff average of 0.1 USD/kWh is more than double Nigeria's average of 0.041 USD/kWh, meaning that Ghanaians

pay much more for the same energy usage. This higher tariff greatly influences the overall energy costs, as seen in previous analyses.

Additionally, Ghana has longer appliance usage hours, particularly for appliances like fridges and freezers, which run for an average of 20.3 hours/day compared to 11.8 hours/day in Nigeria. While the average wattage of appliances in both countries is comparable, this extended usage in Ghana drives up total daily consumption. This effect is even more pronounced in the non-energy-efficient scenario, amplifying energy consumption and costs due to older, less efficient appliances. Consequently, the combination of higher tariffs and longer usage hours explains why Ghana faces significantly higher energy costs than Nigeria.

Annual Electricity Cost and Comparison for Nigeria and Ghana

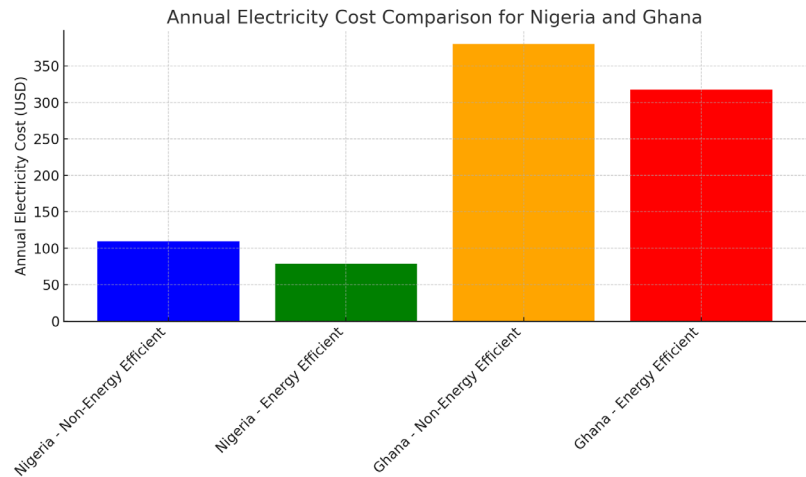


Fig. 20. Annual Electricity Cost and Comparison for Nigeria and Ghana

Total Solar PV and Battery System Cost for Nigeria and Ghana

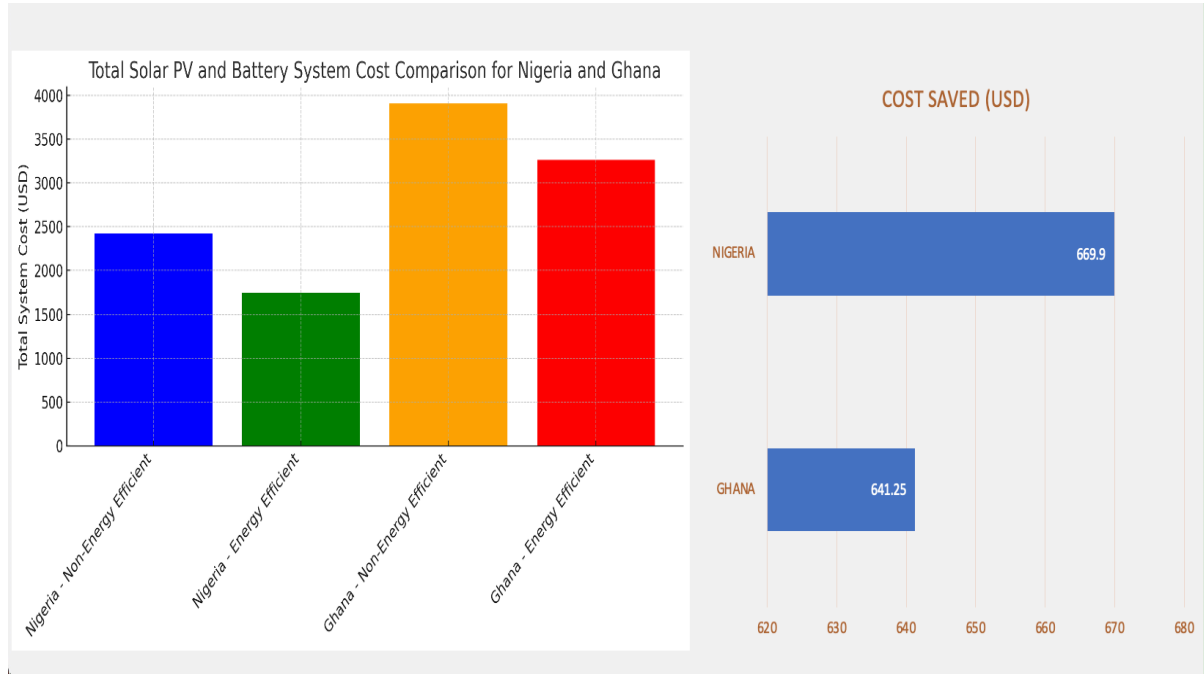


Fig. 21. Total Solar PV and Battery System Cost for Nigeria and Ghana

The left side of the graph shows the total cost of installing solar PV and battery systems for both energy-efficient and non-energy-efficient scenarios in Nigeria and Ghana. The total system cost in Nigeria for non-energy-efficient households is about 2,420 USD, while energy-efficient households require a smaller system, costing around 1,750 USD. In Ghana, the non-energy-efficient system costs significantly more, approximately 3,900 USD, while energy-efficient systems cost around 3,260

USD. This indicates that energy efficiency reduces electricity consumption and lowers the cost of implementing solar systems by requiring smaller solar PV and battery setups.

The right side of the graph highlights the cost savings in each country by comparing the energy-efficient and non-energy-efficient scenarios. In Nigeria, the cost savings from adopting energy-efficient technology amount to about 669.90 USD; in Ghana, the savings are 641.25 USD. These savings come from the reduced need for larger, more expensive solar PV and battery systems in energy-efficient households. Although the savings are similar in both countries, Ghana's overall system costs are higher, reflecting its higher energy consumption and electricity tariff.

LCOE Calculations Under the Scenarios for Nigeria and Ghana

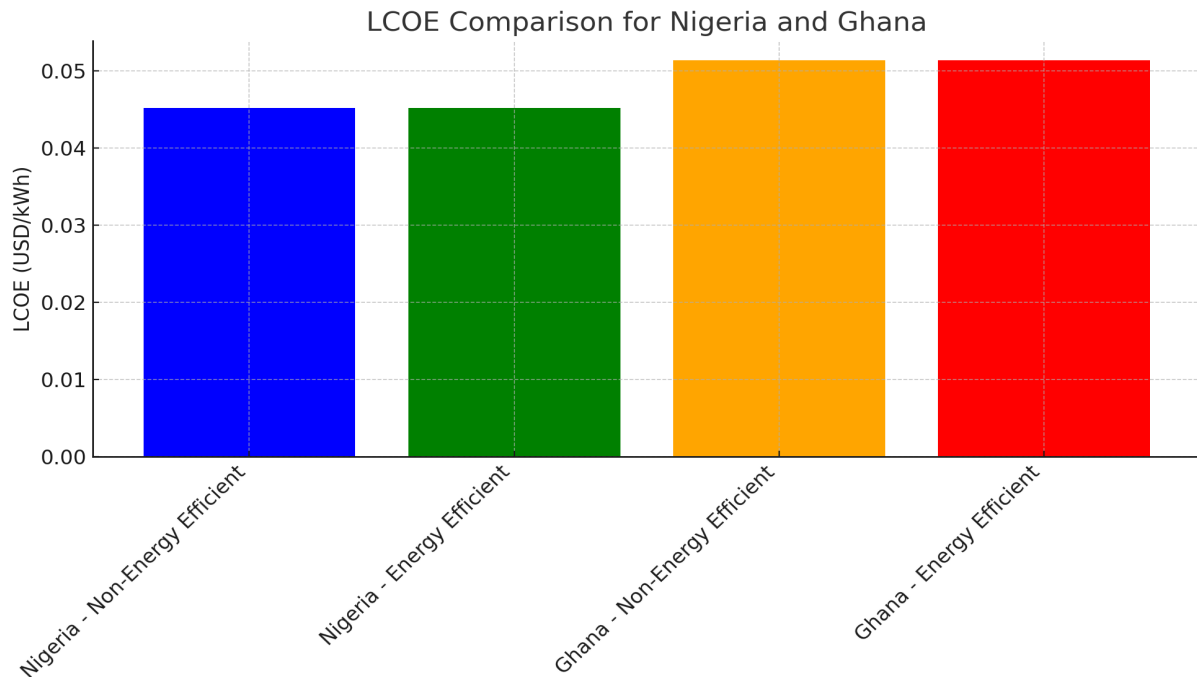


Fig. 22. LCOE Calculations Under the Scenarios for Nigeria and Ghana

The LCOE is relatively close between energy-efficient and non-energy-efficient systems in both countries. Ghana has a slightly higher LCOE due to higher energy consumption and system costs. The graph demonstrates that while energy efficiency reduces the overall cost of installing solar PV systems, the per-unit energy cost remains stable across both scenarios.

NPV Comparison for Solar System Adoption under Analyzed Scenarios Nigeria and Ghana

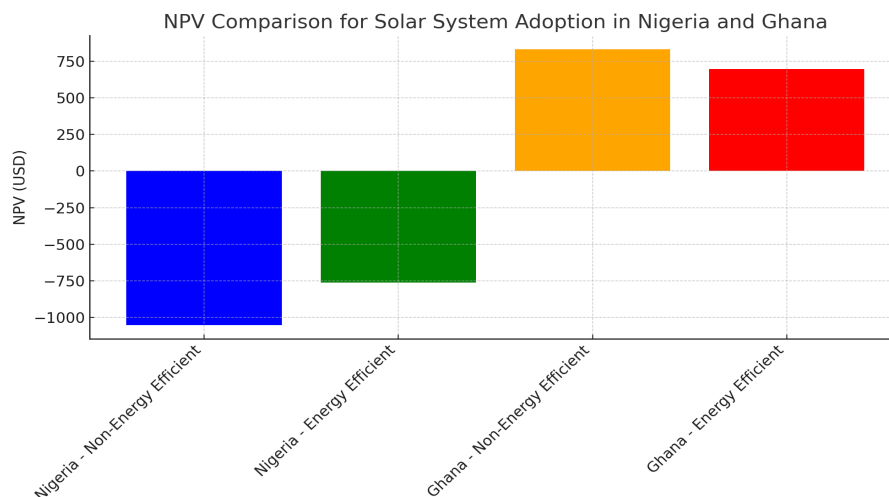


Fig. 23. NPV Comparison for Solar System Adoption in Nigeria and Ghana

Figure 21 explains the Net Present Value (NPV) for solar system adoption in Nigeria and Ghana under non-energy- and energy-efficient scenarios. The NPV measures the profitability of investing in solar PV systems, considering the initial investment costs and the projected savings over the system's lifetime. A positive NPV indicates the investment is profitable, while a negative NPV suggests it may not be financially viable. In Nigeria, both scenarios show negative NPVs, with the non-energy-efficient scenario having a more significant loss (around -1000 USD) than the energy-efficient scenario (around -750 USD). This suggests that under the conditions at the time of this study, adopting solar systems in Nigeria may not be economically viable due to lower electricity tariffs and longer payback periods.

In contrast, the NPVs for Ghana are positive in both scenarios, with the non-energy-efficient scenario showing a higher NPV (around 700 USD) compared to the energy-efficient scenario (around 500 USD). This indicates that solar system adoption is economically viable in Ghana, especially in households with higher energy consumption. The higher electricity tariffs in Ghana make solar investments more attractive as they offer more significant savings over time, resulting in positive returns.

Conclusion

The study found that energy consumption patterns for household appliances such as refrigerators, freezers, and lighting systems significantly vary between Nigeria and Ghana. Ghanaian households generally exhibit higher daily energy consumption, attributed to longer operational hours and higher wattage of appliances compared to their Nigerian counterparts. This difference emphasizes the importance of addressing appliance usage patterns alongside energy efficiency measures.

Adopting energy-efficient appliances offers a notable cost-saving opportunity. In Nigeria and Ghana, switching to energy-efficient refrigerators, freezers, and LED bulbs substantially reduces daily and annual electricity costs. However, the cost-saving impact is more pronounced in Ghana due to its higher electricity tariff, making energy-efficient appliances even more critical for reducing household energy bills.

The study highlights the environmental benefits of energy-efficient technologies. Using tools like RETScreen, the analysis demonstrates that switching from compact fluorescent lamps (CFLs) to LED lighting reduces greenhouse gas emissions significantly. Both Nigeria and Ghana show reductions in emissions equivalent to taking multiple cars and light trucks off the road, emphasizing the potential for residential energy efficiency to contribute to global climate goals.

A comparison of solar PV systems and battery storage for energy-efficient and non-energy-efficient scenarios reveals that energy-efficient households have lower system installation costs. The Levelized Cost of Energy (LCOE) for energy-efficient setups is lower, making the adoption of solar PV systems more economically viable for households that prioritize energy efficiency. The cost savings from solar PV installations are more significant for Ghanaian households due to their higher electricity tariffs.

Appendix

1. The Excel Raw Sheet of the data that was collected, cleaned and analyzed - One drive link – https://docs.google.com/spreadsheets/d/1avBPP9SOAiZPkY5O49t_ao3AsMO5MZonMlOI66_rNs/edit?usp=sharing

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